Lecture 12

Autocratic Politics

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Autocracy: a definition

Definition of democracy:

- Free and competitive legislative elections
- Executive accountable directly or indirectly

Autocracy: any regime which does not satisfy (i) or (ii) (Boix & Svolik, 2013)

No institutional mechanism to hold leaders accountable

Why study autocracy

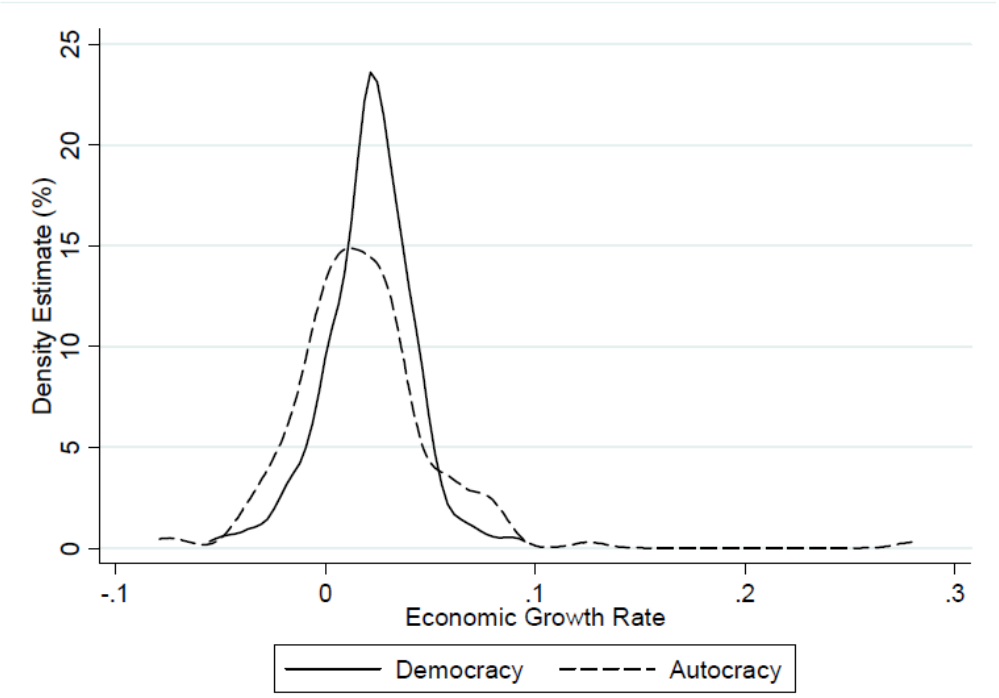
Human history is a history of autocracies

Democracy no longer expanding

Comparative institutions: What do we gain with democracy?

Other puzzle: Very heterogeneous performance among autocracies

Performances: autocracies and democracies



Source: Besley and Kudamatsu (2008)

Autocracy: not unlimited rule

Very few cases of unlimited rule: Stalin, Pol Pot(?), Mao(?)

Autocrats rely on a clique (or selectorate, Bueno de Mesquita et al., 2003)

Members of selectorate can stage a coup

$\Rightarrow$ Impose constraint on autocrat

Autocrats not immune from popular pressure

but cost of replacing autocrats higher for population

Questions

Focus on selectorate (no revolutions):

How do autocrats hold on to power?

How do autocracies' performances compare with democracies'?

Which conditions are favorable to good performance?

Holding on to power: the politics of fear

Padró i Miquel (2007) studies how threat of replacement holds coalition together

Leader can tax activities differently

Leader can provide targeted benefits (e.g., positions in bureaucracy)

Beneficiaries do not want to lose their benefits...

... despite rents extracted by leader

Set-up

Infinitely repeated game

2 groups: A and B of size π_A, $\pi_B = 1 - \pi_A$

Two economic activities: a and b

Group J better at activity j, production/income w

Group J in activity $k \neq j$ gets income: $w - \theta$

Group J starts in activity j, switching decision $z^J \in \{0, 1\}$

Leadership and policy choice

Leader L^S from group S

Start with leader from group A

Leader chooses:

- taxes on activities τ^{Sa} and τ^{Sb}
- targeted transfers to groups $\eta^{SA} \geq 0$ and $\eta^{SB} \geq 0$

Leadership transition

Survival of leader L^S depends on support of group S

With support, survives with probability 1

Without support, leader loses power

New leader from group S with probability γ, from group K with probability $1 - \gamma$

$\Rightarrow$ Group S is selectorate

$\Rightarrow$ Power struggle at the top creates uncertainties

Per-period payoff

Members of group A:

$$U^A = R(\eta^{SA}) + w - (1 - z^A)\tau^{Sa} - z^A(\theta + \tau^{Sb})$$

$R(\cdot)$ is increasing, concave, s.t. $R(0) = 0$

Leader L^S:

$$\begin{aligned} U^{L^S} =& \pi^A((1 - z^A)\tau^{Sa} + z^A\tau^{Sb} - \eta^{SA}) \\ &+ (1 - \pi^A)((1 - z^B)\tau^{Sb} + z^B\tau^{Sa} - \eta^{SB}) \end{aligned}$$

Discount factor: $\delta < 1$

Timing: period t

1. Leader L^S announces taxes and transfers: $P_t = \{\tau_t^{Sa}, \tau_t^{Sb}, \eta_t^{SA}, \eta_t^{SB}\}$

2. Citizen of group S decides whether to support $s_t = 1$ or not $s_t = 0$

3. Economic choices (switching) and activities occur

4a. If $s_t = 1$, then P_t is implemented and L^S retains power in the next period

4b. If $s_t = 0$, then $P^o = (0, 0, 0, 0)$ is implemented, L^S is ousted, and $S_{t+1} = S_t$ with probability γ

Equilibrium concept

Usual problem with this type of games: infinitely many equilibria (similar to Folk Theorem)

Refinement: (Stationary) Markov Perfect Equilibrium (MPE)

Basic idea of MPE:

- strategies depend on previous periods only through last realization of a state variable
- strategies must induce a SPNE in each period

State variable in model? group identity of leader (S)

Implication: strategies will not be indexed by t

Constraints on leader L^A

Leader wants to maximize revenue from taxation $\Rightarrow$ no sector switches:

- $\tau^{Ab} \leq \tau^{Aa} + \theta$
- $\tau^{Aa} \leq \tau^{Ab} + \theta$ (not binding)

Leader wants to keep support from group A:

$$w - \tau^{Aa} + R(\eta^{AA}) + \delta V^A(A) \geq w + \gamma \delta V^A(A) + (1-\gamma)\delta V^A($$

$V^J(S)$: continuation value as a function of the leader's group identity S (part of the equilibrium)

Leader's maximization problem

Using (1), (2) and reasoning above, L^A solves:

$$\max_{\tau^{Aa},\tau^{Ab},\eta^{AA},\eta^{AB}} \pi^A(\tau^{Aa} - \eta^{AA}) + (1-\pi^A)(\tau^{Ab} - \eta^{AB})$$

$$subject\ to\ \tau^{Ab} \leq \tau^{Aa} + \theta$$

$$\tau^{Aa} - R(\eta^{AA}) \leq \delta(1-\gamma)(V^A(A) - V^A(B))$$

Analysis

1. Transfer to group B no impact on survival $\Rightarrow \eta^{AB*} = 0$
2. Tax group B as much as possible $\Rightarrow \tau^{Ab*} = \tau^{Aa*} + \theta$
3. Leader optimizes $\Rightarrow$ (2) holds with equality

$$\tau^{Aa*} = \delta(1-\gamma)(V^A(A) - V^A(B)) + R(\eta^{AA*})$$

4. Substitute the above (recall $\tau^{Aa} \leq \tau^{Ab} + \theta$ does not bind) into leader's problem: $\max_{\eta^{AA}} \Big\{ -\pi^A \eta^{AA} + (1 - \pi^A)\theta + \delta(1-\gamma)(V^A(A) - V^A(B)) + R(\eta^{AA}) \Big\}$

Transfer to group A determined by FOC: $R'(\eta^{AA*}) = \pi^A$

Continuation values

From $V^A(A) = w - \tau^{Aa} + R(\eta^{AA}) + \delta V^A(A)$, obtain

$$V^A(A) = \frac{w - \tau^{Aa*} + R(\eta^{AA*})}{1-\delta}$$

Subgame perfection implies $V^A(B) = w - \tau^{Ba} + \delta V^A(B)$, i.e,

$$V^A(B) = \frac{w - \tau^{Ba*}}{1-\delta} = \frac{w - \tau^{Bb*} - \theta}{1-\delta}$$

Hence

$$V^A(A) - V^A(B) = \frac{\tau^{Bb*} + \theta - \tau^{Aa*} + R(\eta^{AA*})}{1-\delta}$$

and

$$\tau^{Aa*} = \frac{\delta(1-\gamma)}{1-\delta}(\tau^{Bb*} + \theta - \tau^{Aa*} + R(\eta^{AA*})) + R(\eta^{AA*})$$

Continuation values

Using similar reasoning:

$$\tau^{Bb*} = \delta(1-\gamma)(V^B(B) - V^B(A)) + R(\eta^{BB*}),$$

with $R'(\eta^{BB*}) = (1-\pi^A)$

Using the same steps as above we obtain

$$\tau^{Bb*} = \frac{\delta(1-\gamma)}{1-\delta}(\tau^{Aa*} + \theta - \tau^{Bb*} + R(\eta^{BB*})) + R(\eta^{BB*})$$

Last step: system of two equations (3)-(4) gives (τ^{Aa*}, τ^{Bb*})

Two remarks on policy choices

$$\tau^{Aa*} = \delta(1-\gamma)(V^A(A) - V^A(B)) + R(\eta^{AA*})$$

(i) $\tau^{Aa*} > R(\eta^{AA*})$: Leader extracts rents from supporters

(ii) $\tau^{Aa*} - R(\eta^{AA*})$ decreasing with γ

$\Rightarrow$ Less rents when selectorate is secure

Limits on $\tau^{Aa*} \Rightarrow$ limits on τ^{Ab*}: spillover from A to B due to labor mobility

But why doesn't leader take everything?

What constrains leader from taking everything?

Need support from group A

But if group A gets nothing from $L^B \Rightarrow$ continuation value from not supporting is 0

$\Rightarrow$ Leader A does not need to make concessions

Very important subtle assumption: rebellion $\Rightarrow$ no taxation

Guarantee minimal income to A (see (2)) and makes model work...

Including selection

Padró i Miquel (2007) shows that secure selectorate can reduce rent seeking

Besley & Kudamatsu (2008) include problem of adverse selection

- Leader can be corrupt or honest
- Selectorate can trigger power contest by ousting leader

Same result: As selectorate becomes more secure, corrupt leader extracts less rents

Secure selectorate *also* improves selection of leaders

Set-up

Two periods $t \in \{1, 2\}$

Two groups: A, size β and B, size $1 - \beta$

Leader representative of one of the two groups makes two decisions:

- General interest: $e_t \in \{0, 1\}$
- Targeted transfers: $\sigma_{At} + \sigma_{Bt} \leq 1,\ \sigma_{Jt} \geq \underline{\sigma}$

Effect of general interest policy depends on $\omega_t \in \{0, 1\}$

Prior: $Pr(\omega_t = 1) = 1/2$

Population

Both groups want general interest policy to match state

Care about targeted transfer as well

Group J's per-period utility:

$$U_J^t = \Delta \mathbb{I}_{\{e_t = \omega_t\}} + \sigma_{Jt}$$

Population subdivided between franchised ($n < 1$) and unfranchised

Leadership

Leader can be corrupt (c, prob. $1-\pi$) or honest (h, prob. π): $\tau \in \{c, h\}$

Leader gets utility Δ if $e_t = \omega_t$

Corrupt leader gets rents r_t if $e_t \neq \omega_t$

r_t drawn according to $G(\cdot)$ i.i.d., with $G(\Delta) = 0$ and $E(r) = \bar{r}$

(Way to deal with mixed strategy)

The utility of leader $L \in \{A, B\}$ when in power:

$$V_L(e_t, \sigma_t; \tau) = \Delta \mathbb{I}_{\{e_t = \omega_t\}} + \sigma_{Jt} + r \mathbb{I}_{\{e_t \neq \omega_t, \tau = c\}}$$

Key assumption: When not in power, same utility as (5)

Political transition (1)

Leader A in power in period 1

Political transition in two stages

(i) Enfranchised voters from group A (selectorate) decides whether to oust leader

If keep leader, leader A in power in period 2

If oust leader, stage 2 of transition

Political transition (2)

(ii) Political contest between random (new) leader from A and B

"Probabilistic voting" approach

Probability group A wins contest depends on:

- proportion of selectorate favouring A: κ (exogenous)
- shock on B's strength: $\eta \sim U[-1/2, 1/2]$

Probability group A wins:

$$Pr((1-\kappa)+\eta \leq \kappa) := \gamma(\kappa) = \begin{cases} 1 & \text{if } \kappa > 3/4 \\ 2\kappa - 1/2 & \text{if } 1/4 \leq \kappa \leq 3/ \\ 0 & \text{if } \kappa < 1/4 \end{cases}$$

Timing

Period 1:

1. Nature determines ω_1, r_1, and type of leader A in power in period 1

2. Leader A observes his type, r_1, and ω_1, chooses $e_1, \sigma_{A1}, \sigma_{A2}$

3. Citizens observe first-period payoff. Selectorate decides whether to keep leader. If keep leader, no political transition. If oust leader, political transition

Period 2.

Stages 1-2 repeated

Equilibrium concept: PBE

Equilibrium: Period 2 choice

No risk of being replaced $\Rightarrow$ leader maximizes his period 2 utility

- Honest leader chooses $e_2 = \omega_2$
- Corrupt leader chooses $e_2 \neq \omega_2$

Leader always favors his own group: $\sigma_{L2} = 1 - \underline{\sigma}$

Selectorate decision

Denote $\mu(U_A^1)$ the selectorate's posterior that leader is honest as a function of first-period payoff

If selectorate keeps leader, gets: $\mu(U_1^A)\Delta + 1 - \underline{\sigma}$

If selectorate ousts leader, gets: $\pi\Delta + \gamma(\kappa)(1 - \underline{\sigma}) + (1 - \gamma(\kappa))\underline{\sigma}$

If $\mu(U_A^1) \geq \pi$, then never ousts leader

If $\mu(U_A^1) < \pi$, then ousts leader iff:

$$(\pi - \mu(U_A^1))\Delta > (1 - \gamma(\kappa))(1 - 2\underline{\sigma})$$

Selectorate decision (2)

$$(\pi - \mu(U_A^1))\Delta > (1 - \gamma(\kappa))(1 - 2\underline{\sigma})$$

Suppose $\pi\Delta < 1 - 2\underline{\sigma}$, two interesting cases

(i) $\gamma(\kappa) = 0 \Leftrightarrow$ personal rule (e.g., Franco in Spain, Marcos in Philippines)

Selectorate' strategy? Never replace corrupt leader ($\mu = 0$)

(ii) $\gamma(\kappa) = 1 \Leftrightarrow$ entrenched autocracy (e.g., CCP in China)

Selectorate' strategy? Always replace whenevr $\pi > \mu(U_A^1)$

Leader's first-period decision

Denote $\rho(U_A^1)$ probability leader is ousted as a function of first-period payoff

Honest leader always produces payoff $\Delta + 1 - \underline{\sigma}$ (i.e., $e_1 = \omega_1$)

$\Rightarrow \mu(\Delta + +1 - \underline{\sigma}) \geq \pi \Rightarrow$ selectorate never ousts after $U_A^1 = \Delta + 1 - \underline{\sigma}$

$\Rightarrow \mu(1 - \underline{\sigma}) = 0 \Rightarrow$ Corrupt leader's decision depends on $\rho(1 - \underline{\sigma})$

Corrupt leader's first-period decision

If $\rho(1-\underline{\sigma}) = 0$ (i.e., $\gamma(\kappa)$ is low), then no risk of ousting

$\Rightarrow$ Corrupt leader chooses $e_1 \neq \omega_1$

$\Rightarrow$ Weak selectorate: Bad policy choice, no selection

If $\rho(1-\underline{\sigma}) = 1$ (i.e., $\gamma(\kappa)$ is high), trade-off for corrupt leader

Choose right policy if and only if:

$$r \leq \mathbb{E}\{r\} + (1-\pi)\Delta + (1-\gamma(\kappa))(1-2\underline{\sigma}) \equiv R$$

$\Rightarrow$ Strong selectorate: Less corruption (probability $1 - G(R)$ decreasing in γ), some selection

Heterogeneous performance by autocracies

Besley & Kudamatsu (2008) explain heterogeneous performances by autocracies

Cause: Strength (independence) of selectorate

Strong selectorate $\Rightarrow$ High growth, some selection, little corruption

Weak selectorate $\Rightarrow$ Low growth, no selection, high corruption

Model can also be used to compare autocracies and democracies

Comparison with democracy

Democracy same game as above except for stage 3 in period 1

3. Citizens observe first-period payoff. Party A decides whether to keep incumbent.

If keep incumbent, incumbent faces random challenger from party B.

If oust incumbent, random party A member faces random challenger from party B.

Big differences:

- Election always happens
- Everybody votes (probability group A wins election is $\gamma(\beta)$)

Comparison with democracy (2)

Party leaders keep incumbent iff $\mu(U_A^1) \geq \pi \Leftrightarrow U_A^1 = \Delta$

Honest incumbent always chooses $e_1 = \omega_1$

Corrupt incumbent chooses $e_1 = \omega_1$ if and only if:

$$r \leq \gamma(\beta)\bar{r} + (1-\pi)\Delta \equiv D \leq R$$

Democracies more homogeneous: No dependence on party's strength

Democracy is associated with more corruption and more selection

Transfers are also more equitable (over time) in democracy

Are there differences between democracies and auto

Leaders in both regimes face constraints

But constraints are different:

- Transition irregular in autocracies, depends on strength of selectorate
- Transition regular in democracies

$\Rightarrow$ Some, but moderate consequences for policy choices according to Besley & Kudamatsu (2008)

What do the data say?

Political regimes and policy choices

Mulligan et al. (2004) study how regimes affect policies

Contrast political economy view (institutions matter, e.g. Besley) and economics view (efficiency matters, e.g. Becker)

Using cross-country regressions, show:

- Institutions have no effect on economic policies: expenditures, taxes (except income taxes)
- Institutions have effect on political policies: torture, civil and religious liberties

Problem? Institutions are sticky so no fixed effect

Political regimes and policy choices (2)

To understand effect of institutions, need institutional changes within country

But not enough, need non-homogeneous reform

Martinez-Bravo et al. (2014) study staggered adoption of democratic elections in Chinese villages

Study how reform affects public good provision and officials retention

Political reform in China

Before reform, village executive constituted of:

- CCP representative
- Village Committee members appointed by CCP county-level government

After reform, village executive constituted of:

- CCP representative
- Village Committee (VC) members elected by villagers

Authors show more decisions by VC after reform

Identification strategy

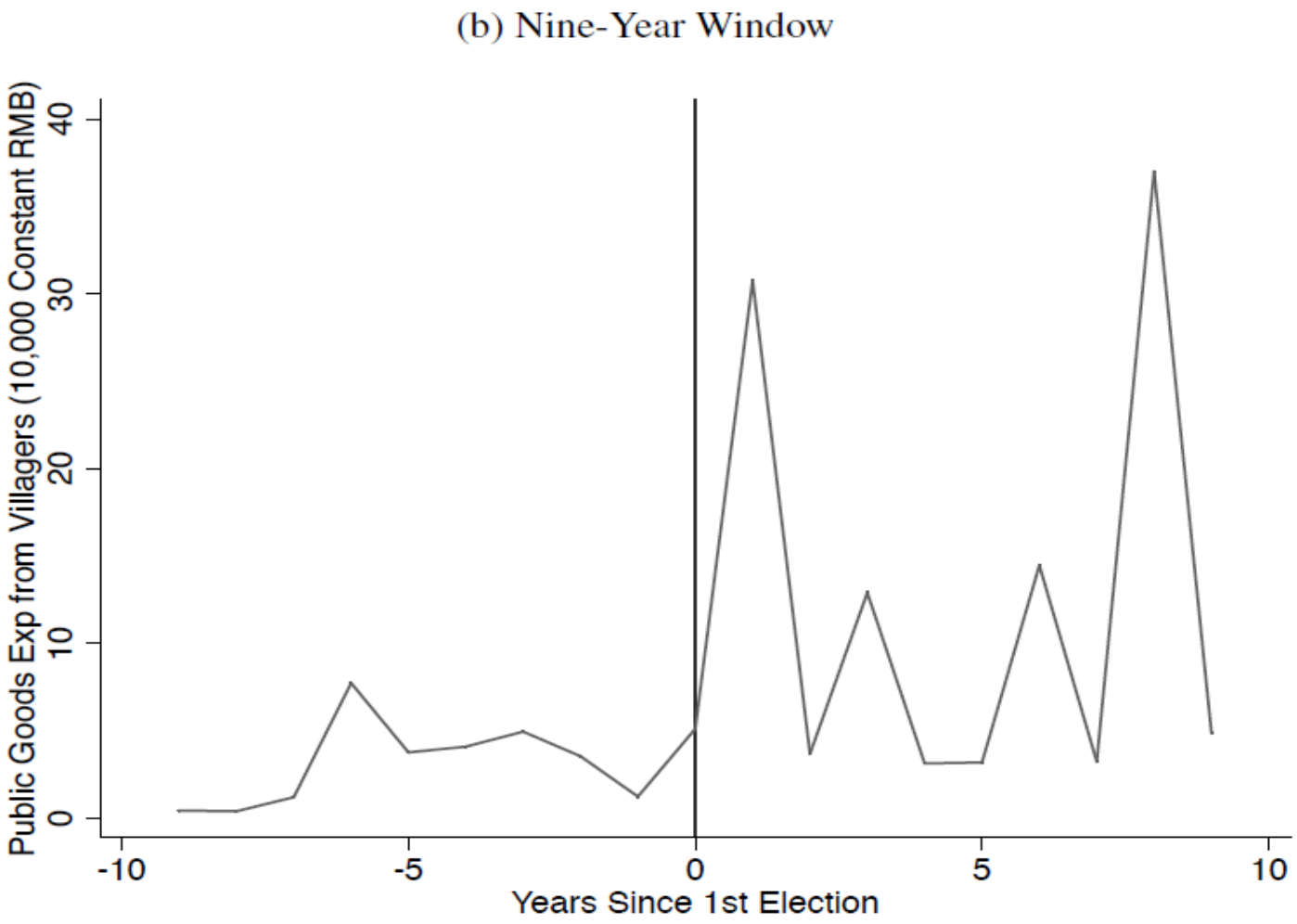
Reform not adapted at the same time:

- Across provinces (not random)
- Within province (quasi-random)

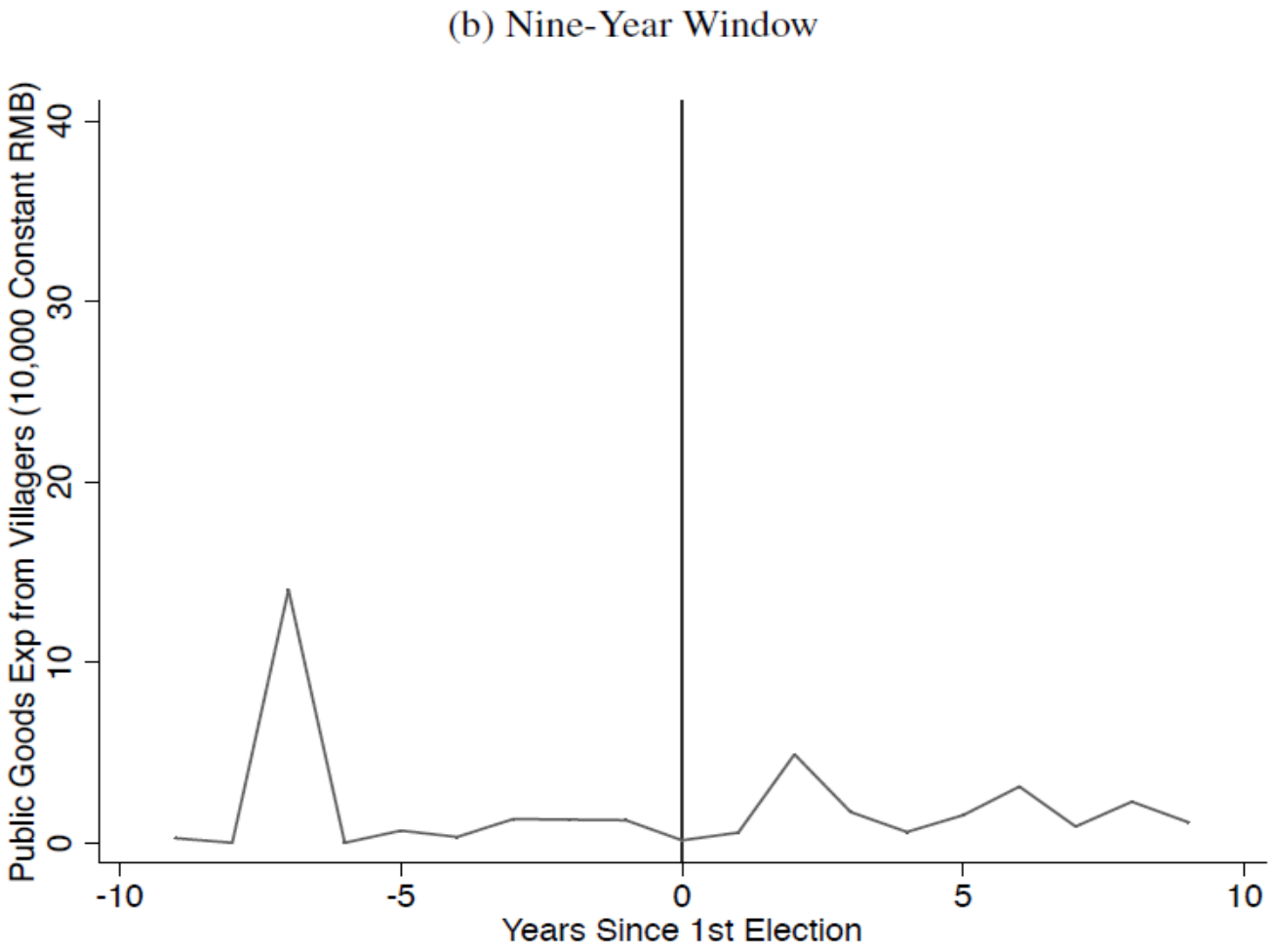
Identification comes from within province variations (province fixed effect)

Use a random sample of Chinese villages (collect data on these villages)

Public good expenditures by villages



Public good expenditures by upper-government



Regression results

	Dependent Variable					
	From All Sources	From Villagers	From Other Sources	From All Sources	From Villagers	From Other Sources
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Public Goods Expenditures (Constant 10,000 RMB)					
Dependent Variable Mean	*13.81*	*9.46*	*4.28*	*13.81*	*9.46*	*4.28*
Post 1st Election	18.401	16.351	2.165	15.311	16.080	-0.672
	(9.247)	(9.433)	(2.029)	(7.190)	(7.717)	(1.525)
Wild Bootstrap p-value	[0.086]	[0.148]	[0.384]	[0.042]	[0.046]	[0.738]
Controls:						
Province - Trend	N	N	N	Y	Y	Y
Observations	4,340	4,340	4,340	4,340	4,340	4,340
R-squared	0.108	0.097	0.067	0.113	0.103	0.073
Clusters	29	29	29	29	29	29

Political turnover

	Dependent Variable: Characteristic of Village Leader			
	Turnover	Age	Years of Edu	Party Member
	(1)	(2)	(3)	(4)
	A. Village Chairman (VC)			
Dependent Variable Mean	*0.16*	*42.15*	*7.88*	*0.77*
Post 1st Election	0.045	-2.442	0.791	-0.034
	(0.020)	(1.037)	(0.224)	(0.043)
Wild Bootstrap p-value	[0.001]	[0.040]	[0.000]	[0.468]
Obs	4,312	4,188	4,194	4,274
R^2	0.065	0.430	0.611	0.484
	B. Party Secretary (PS)			
Dependent Variable Mean	*0.15*	*44.35*	*8.42*	
Post 1st Election	-0.006	0.312	-0.121	
	(0.018)	(0.632)	(0.139)	
Wild Bootstrap p-value	[0.199]	[0.607]	[0.478]	
Obs	4,365	3,546	4,356	
R^2	0.071	0.677	0.572	

Electoral incentives

Compare public good provisions of incumbent VC who remained in office after first election

	Total, In Levels (1)
Post 1st Election	54.509
	(28.762)
Wild Bootstrap p-value	[0.432]
Post 1st Election x 1st Election VC Change	-40.826
	(28.515)
Wild Bootstrap p-value	[0.548]
Observations	4,340
R^2	0.110
Clusters	29

Autocratic politics: other interesting aspects

Autocrats do not govern alone, need ministers (viziers): Egorov and Sonin (2011)

Autocrats replaced if does not take precautionary measures and enemy is strong

Viziers analyze threats to autocrats, make recommendations, and risk being punished if mistake

But punishment conditional on autocrat surviving

Competence $=$ high ability to spot strong enemy

$\Rightarrow$ If lie about strength, autocrat fails

$\Rightarrow$ Hard to punish $\Rightarrow$ easy to bribe for strong enemy

$\Rightarrow$ Autocrats prefer non-competent viziers

Autocratic politics: other interesting aspects

Svolik (2009) considers how autocrats extend their power

Moral hazard associated with autocracy:

- Autocrats try to divert resources to increase their hold on power
- Selectorate unable to perfectly monitor autocrats' actions

Selectorate can stage a coup to limit autocrats' power; success depends on autocrats' power

Over time, probability of diversion and coup increases until autocrats hold on power is secure

Summary

Autocracy $\not\Rightarrow$ no accountability of leader

Degree of accountability depends on selectorate's incentives to tolerate autocrat's misbehaviour

Less tolerance if selectorate is strong/secure

Can explain heterogeneous performances across autocracies

But selectorate not a random sample of population

"Autocratic accountability" good for unfranchized citizens if and only if convergence of interests with selectorate

The simplified AR model

Society is divided in two groups: $e$lite and $m$ajority

Infinitely repeated interaction, time indexed by t

Each group acts as a unitary actor

Choices:

- $x_t \in [0, 1]$: proposed share of surplus going to m
- $\rho_t \in [0, 1]$: degree of political liberalization
- r_t: choice of revolt by m

Stochastic events:

- $\mu_t \in \{k, 1\}$, permanent surplus destruction post-revolution (if there is one)
- $\alpha_t \in \{m, e\}$, proposer of x_t ($\Pr(\alpha_t = m) = \rho_t$)

Parameters and Assumptions

$q = \Pr(\mu_t = k)$: probability of future unrest

$\delta \in (0, 1)$: discount factor

k: cost of unrest

Also:

- Game starts with ρ_0
- First changes to ρ_t is permanent and irreversible
- Revolt yields to permanent destruction of μ_t
- m keeps all remaining surplus after revolt

Timing

- μ_t realized
- if $\rho_{t-1} = 0$, e chooses to liberalize ($\rho_t > 0$) or not ($\rho_t = 0$)
- if $\rho_t = 0$, e chooses x_t
- if $\rho_t > 0$ $\alpha \in \{e, m\}$ chooses x_t with $\Pr(\alpha = m) = \rho_t$
- m decides whether to revolt (gets $1 - \mu_t$) or not (gets x_t)

Assumptions

1. $\delta > k$ (revolution cannot be always avoided):

$$\frac{1-k}{1-\delta} > 1$$

2. $q < q^* = \frac{\delta - k}{\delta}$ (full liberalization avoids revolution):

$$1 + \delta q \frac{1}{1-\delta} > \frac{1-k}{1-\delta}$$

3. $\delta + k < 1$ (just to simplify the math)

Analysis

Equilibrium Concept: Stationary Markov Perfect Equilibrium

State depends on ρ, μ, α (only matters if $\rho > 0$)

First, notice that $x(0, 1, \alpha) = 0$ (when not threatened, e keeps everything)

Construct an eqm in which e liberalizes to ρ' whenever $\mu = k$

Four relevant states (ρ', μ, α):

- $x(\rho', k, m) = 1$ (when in power, m keeps everything)
- $x(\rho', 1, m) = 1$ (when in power, m keeps everything)
- $x(\rho', 1, e) = 0$ (when not threatened, e keeps everything)
- $x(\rho', k, e) = \tilde{x}(\rho')$ (we need to characterize it)

Characterizing $\tilde{x}(\rho')$

1. e needs to avoid a revolution:

$$V_m(\rho', k, e) = \frac{1-k}{1-\delta}$$

2. $V_m(\rho', k, e)$ is the value of accepting $\tilde{x}(\rho')$:

$$V_m(\rho', k, e) = \tilde{x}(\rho') + \delta V_m(\rho')$$

where

Characterizing $\tilde{x}(\rho')$ - part 2

Substituting yields

$$\tilde{x} + \frac{q(1-\rho')\tilde{x} + \rho'}{1-\delta} = \frac{1-k}{1-\delta}$$

requires (the former ensures that the latter is ≤ 1)

$$\rho' \geq \frac{\delta(1-q)-k}{\delta(1-q)}$$

$$\tilde{x}(\rho') = \frac{1-k-\delta\rho'}{1-\delta(1-q)-\delta q\rho'}$$

Notice that $\tilde{x}(\rho')$ is decreasing in ρ'

- Trade-off between redistribution and political power

Optimal degree of liberalization

Value to the elite

$$V(\rho') = (1-\rho')[1-q+q\tilde{x}(\rho')]$$

strictly decreasing in ρ'

As a result, equilibrium liberalization is

$$\rho^*(k) = \frac{\delta(1-q)-k}{\delta(1-q)}$$

$$\rho^*(1) = 0$$

Observe that equilibrium liberalization decreases in q

Higher threat reduces extension of political rights